

INTELLIGENT RUBRICS - BASED EVALUATION SYSTEM FOR ALGORITHMS PSEUDOCODES & FLOWCHARTS

SHANMUGA PRIYA G
Assistant Professor
Department of Computer
science and Business Systems
Dr. N. G. P Institute of
Technology
Coimbatore, India
shanmugapriya.g@drngpit.ac.in

NITHYA SRI J
Department of
Computer science and
Business Systems
Dr. N. G. P Institute of
Technology
Coimbatore, India
23cb037@drngpit.ac.in

POOJA G
Department of
Computer science and Business
Systems
Dr. N. G. P Institute of
Technology
Coimbatore, India
23cb038@drngpit.ac.in

SASMITA C J
Department of
Computer science and Business
Systems
Dr. N. G. P Institute of
Technology
Coimbatore, India
23cb047@drngpit.ac.in

Abstract - The evaluation of flowcharts, algorithms, and pseudocode plays a crucial role in Computer Science education; however, manual assessment methods are often time-consuming, inconsistent, and prone to subjective interpretation. To address these limitations, this paper presents an Intelligent Rubric-Based Evaluation System that automates the grading process using artificial intelligence techniques. The proposed system applies predefined rubrics to ensure consistent, transparent, and objective assessment of student submissions. A user-friendly interface is developed using Streamlit, enabling students to submit algorithms, flowcharts, and pseudocode either as text or images. The backend, implemented in Python, manages evaluation logic and workflow orchestration. All submissions, evaluation criteria, scores, and feedback are securely stored in a relational SQL database for structured data management and performance tracking. For image-based submissions, Optical Character Recognition (OCR) technology is used to extract textual and structural information from handwritten or drawn content. The extracted data is then analyzed using rubric-guided semantic reasoning combined with pre-trained transformer inference models to evaluate correctness, logical flow, clarity, and completeness. The proposed system significantly reduces manual grading effort while enhancing fairness, reliability, and scalability in digital assessments. By providing instant, detailed feedback, it supports continuous learning and improves students' computational thinking skills. The system demonstrates an effective approach to implementing AI-driven, scalable, and transparent evaluation methods in modern educational environments.

Keywords: *Assessment Rubrics, Optical Character Readers (OCR) and Transformers (used in*

Document Analysis), Charting Algorithms, Flowcharting, Evaluating Systems (FES), Pseudo Code, Stream Lit (Stream) and Artificial Intelligence in Education (AIE).

I. Introduction

In the field of computer science education, students learn logical reasoning and problem-solving techniques by learning how to create flowcharts, algorithms and pseudocode, because they allow students to visualize the process by which a problem is solved sequentially from the beginning to the end. It is essential for educators to assess their students in this area, so that they may become adequately trained in the areas of computer science. The traditional method for evaluating programming by tutors is labour intensive and can produce enormous amounts of workload from thousands of students, therefore introducing ambiguity, differing opinions, as well as mistakes, when multiple tutors are involved in grading flowchart or algorithm submissions.

With the introduction of artificial intelligence, automated education evaluation systems have received a great deal of attention; particularly since Optical Character Recognition (OCR), Natural Language Processing (NLP) and transformers have allowed users to evaluate handwritten and image-based submissions, and those evaluations can now be performed objectively and transparently, using well defined evaluation rubrics. The intelligent rubrics-based evaluator assists to evaluate a given submission for flowcharts, algorithms, and pseudo-code using an automated way of evaluating by a computer programme. This computer application is designed using a back-end Java Implementation, which communicates with a SQL database. The submissions to be evaluated are through image submissions, using Optical Character Recognition

(OCR) technologies and the evaluation will use a transformer inference model based on a pre-defined rubric.

This will be able to determine the extent to which a submission carries a logical flow depending upon the definitions prepared for these components.

II. LITERATURE SURVEY

An automated marking system using the evaluation of rules within programming logic has been designed by Smith et al. (2021).[1] The primary emphasis has been given on syntactical correctness. The drawback of this technique is that the knowledge of logic and problem-solving abilities are limited to the current situation. The primary concern of the researcher is to keep emphasis only on developing intelligent systems that are able to learn semantics but not syntactical components. The concern of the current study lies with the needs for reasoning satisfied by NLP evaluation systems.

Kumar & Rao (2020) described the design for an OCR-based evaluation system for the programming answers written by hand.[2] Though the responses given were extracted by scanning the papers of Kumar & Rao, they had difficulties concerning the graphic representation as well as the usage of the flowcharts. The authors were convinced that the employment of both OCR and meaning analysis can play a critical part within the improvement of the accuracy rate.

Li et al. (2022) studied the use of Natural Language Processing techniques in the assessment of pseudocode submitted.[3] The proposed model assessed the control structure and logical flow of submitted code through dependency parsing. Although it was practical as far as typing was concerned, it did not permit image and handwriting submissions. This reiterates the importance of multimodal assessment approaches.

Authors Brown et al. (2019) also suggested a framework for assessing algorithms through rubrics for online learning platforms.[4] They were able to demonstrate the benefits of rubric-based grading in increasing efficacy when compared to manual grading. This work is connected to my proposal since it suggests the completion of rubric-based assessments using fully automated approaches.

In 2023, the model used was the transformer model to analyze the solutions given by students on the basis of coding exercises conducted using the transformer model. [5] For the current discussion, the capacity to compare the models to the previous model on machine models gave a better interpretation of errors. The shortcoming of the current discussion in the application of the model is that the discussion only deals with the model constructed from the text without the usage of flow diagrams and the logic diagram. This further supports the statement that perhaps the transformer inference process could possibly be the most efficient method available.

Patel et al. (2021) have put forward their formulation regarding the approach for the grading system in algorithms based on AI.[6] The authors have stated that the process regarding the articulation of the requirements for correctness and the execution of those requirements in methodology itself takes account of logical rules. The authors have mentioned the issue regarding scalability in case the number of students is quite large for grading.

Nguyen et al. suggested that the topic for research might be the ways of interpretation of flowcharts using the strength of computer vision.[7] Having conducted the research on the topic of flowcharts, the authors were able to achieve some success in recognition tasks but failed to include the logical reasoning among the relevant nodes. The idea suggested by the authors to apply logical reasoning along with the recognition of the shapes using vision is quite applicable to the discussion.

A tool for assessment oriented for the evaluation of answers by using the ability of deep learning for the evaluation of the answer in handwriting was proposed in a paper by Singh et al. (2022).[8] This improved the accuracy level for the evaluation of answers and recognition from the image using image processing. It is clear from the cited paper that accuracy in evaluation and rubric requirements are essential in achieving unbiased results in evaluation.

Chen et al. (2021) studied the role of automated feedback systems using AI models in programming-based learning.[9] Results showed the impact of feedback in learning outcomes, which was largely dependent on the criteria used in the evaluation logic. This suggests the need for intelligent feedback generation based on rubric once more.

Alam and Yusuf designed a digital assessment tool using AI and a database. The designed system will increase grading speed and will not put a lot of pressure on lecturers too.[10] The authors postulated that future systems should be able to process different sources such as images and graphs. The paper is a direct reaction to the aim of the intelligent assessment tool.

III. PROPOSED SYSTEM

The intelligent evaluative system proposed here will automate the evaluation process of flowchart, algorithm and pseudocode submissions through artificial intelligence. This system supports several different types of input forms (e.g., digital files, images, handwritten) whereby images will have text and other non-verbal information extracted via Optical Character Recognition (OCR) and algorithms/pseudocode interpreted using Natural Language Processing (NLP) techniques. For flowcharts, the evaluating system will identify and verify all the standard symbols and the logical connectors between them.

The Evaluating Engine will evaluate the extracted content against rubric criteria (e.g., correctness; logical

flow; clarity; structural completeness) to produce an automatic mark and feedback for learners. Both the submission and results from using the evaluations systems will be accessible via a web-based interface using Streamlit or Flask and allow educators to create/manage rubrics, assess student work, and have students upload their submissions and instantly get back their results. The required structures for ethical records management and technical/globally scalable databases will ensure secure storage/data retention via SQL databases. Finally, with the integration of OCR, NLP, and rubric guided reasoning into the Evaluation Engine will create a system that provides speed, fairness and scale to assess students' computational learning tasks with a significant reduction.

SYSTEM ARCHITECTURE

The architecture design includes five layers that each provide an automated way of evaluating student entry forms for their submissions of Flowcharts, Algorithms, Pseudocode, etc. Each layer has its own function, and thus provides a way to automate the process while providing an intelligent and more accurate evaluation of student submissions, with reduced resources from humans and better transparency.

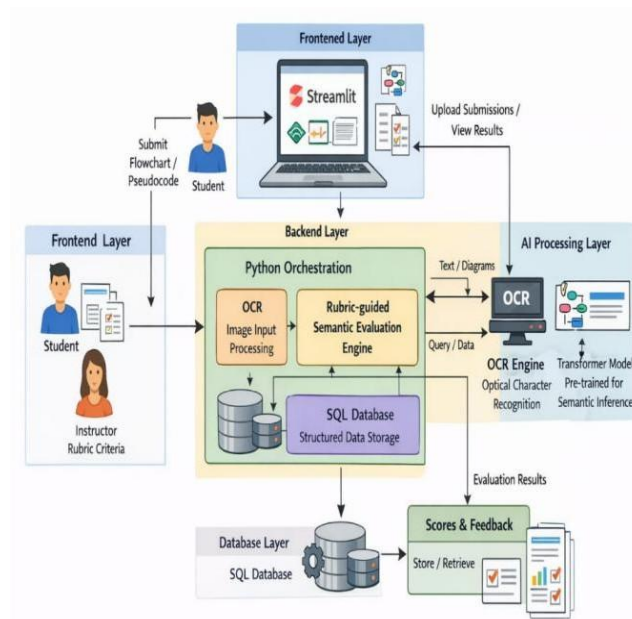


Figure 3.1 System Architecture

A. User Interaction Layer

- The User Interaction layer allows students and instructors to communicate through the platform.
- A Streamlit Web UI allows users to upload handwritten or digital forms (Flowchart/Pseudocode) to the platform.

- Instructors may upload rubrics to help evaluate how well students did on their submissions.
- When students submit their work, the students receive a scored response, along with their feedback.

Responsibilities:

- ✓ Collect student work
- ✓ Authenticate/encrypt users of the system
- ✓ Provide results for student submissions

B. Data Acquisition & OCR Layer

- After uploading an image, it goes through a preprocessing phase to convert it into a text/image.
- Post processed images are enhanced and cleaned of any noise.
- The Optical Character Recognition software (OCR) engine (e.g. Tesseract) recognizes the Base text of an image and extracts it back to a form of digital format (e.g. Text).
- Using Computer Vision technology, a flowchart is identified for identifying structural components.

C. AI Reasoning Layer (Evaluation Engine)

- The AI Reasoning layer is the core of the evaluation process.
- The Natural Language Processing (NLP) model assesses the readiness of algorithms, logical steps, and correctness of algorithms by applying Text Comparison techniques (e.g. T3) and compares and contrasts the steps and rules of submission against Rubrics.
- The Transformer-based Semantic Scorers are used to validate the correctness of each step of an algorithm.
- This layer also ensures students' work will be scored based on the requirements of Rubrics.

D. Structured Database Layer

The information gathered in both Evaluation and Extraction will all be safely contained in an SQL database.

Data and Information will cover:

- Student Data
- Assignment Submission Data
- Rubric Template Data
- Evaluation Result and Data Analytics.

E. Report Generation & Feedback Layer

This layer addresses the generation and delivery of results produced by the Evaluation Process.

Chief Functions and Responsibilities Include:

- Computation of Marks on Rubric Mapping
- Feedback Generation Based on Errors
- Export to PDF & Screen Display
- Progress Dashboard for the Instructor Key

Functions and Responsibility Include:

- Automated Scoring System
- Simple Representation of Results

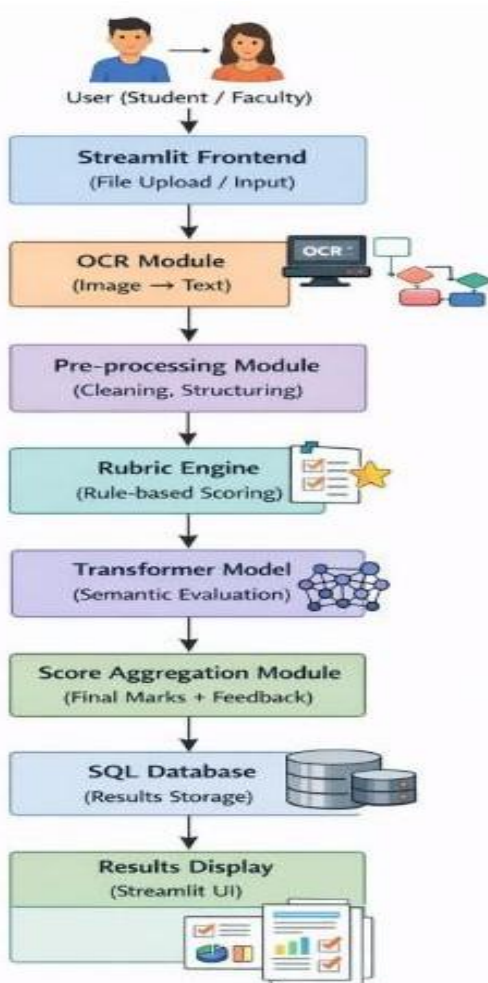


Figure 3.2 Component Flow

COMPONENT FLOW

A. submission/user interaction

- Student logs in to the web interface.
- Student uploads an image of their handwritten or digital Flowcharts, Pseudocode or Algorithms.
- Instructor uploads the rubric criteria against which the student's submission will be evaluated.

- The student's submission is then sent to the OCR and AI processing components of the system.

B. OCR/Image Analysis

- The images uploaded by the student go through a series of preprocessing steps which include Grayscale Image Conversion, Binarization, and Noise Removal.
- From the uploaded algorithms and pseudocode, the OCR extracts any textual elements.
- The Computer Vision technology of the OCR extracts all of the shapes in the flowchart:
 - ✓ Start/Stop
 - ✓ Process Box
 - ✓ Input/Output
 - ✓ Decision Diamond
 - ✓ Flow Connectors
- The OCR outputs a structured view of the flowchart shapes to the Evaluation Engine.

C. AI Decision Support Engine

The intelligent engine reasoning component supports decision making based on the information input into the system. The intelligent engine checks that the order of events makes sense properly (based on control flow, logical flow, etc.) the components of the system check that every step of the process is written according to proper programming syntax for each component; the components, and not one but all of them, check that every component of each student's algorithm has been created using the specified criteria (the rubric), and provide each student a fair and equitable final score.

D. Rubric-Based Scoring Engine

Rubric-Based Scoring Engine – how we gauge the current effectiveness of the algorithm:

- Validity and Completeness of Algorithm.
- Adherence to Flowchart Standards.
- Complexity/Accuracy of Logic and Branching.
- Quality of Problem-Solving Process.
- Semantic Scores based on Natural Language Processing (NLP).

The Rubric-Based Scoring Engine compares the output of a student's algorithm with an instructor's grading rubric as defined by each step/criterion of the grading rubric. The Rubric-Based Scoring Engine assigns total scores for each section of the rubric.

E. SQL Database Management

Database Contents:

- Student Profiles & Submission Logs
- Rubric Templates
- Evaluation Results
- Extracted Features & Recognized Text
- Enables Tracking Progress and Allows Revisiting of an Evaluation If Required

F. Delivery of Results and Feedback

The System Will Provide the Users with The Following:

- ✓ Scores
- ✓ Errors
- ✓ Suggested Corrections
- The results will be visible on the User's Interface and Can Also Be Exported as a PDF File.
- The Instructor Will Have The Ability To Review and Override The Results, When Necessary.

IV. IMPLEMENTATION

The Intelligent Rubrics-Based Evaluator for Flowcharts, Algorithms, and Pseudocode has been developed with a modular software design, making it scalable. The back-end functions of the Evaluator are programmed using the Python programming language. The front-end environment for the interface has been developed using the stream lit web framework. The inputs using images are from Optical Character Recognition (OCR) software that has the capability to translate images to text where the user has the ability to input images from any type of image, whether digital or handwritten. The intention here after uploading the image is that the uploaded file will first be verified if it is indeed one of the supported file formats before it is submitted to the OCR module where the processing will undergo optical character recognition. The texts that are extracted will then proceed to various stages of pre-processing such as noise removal and other processes that will eventually ensure the texts extracted from the images are clean.

The processed text will then be submitted to the AI Analysis Module where both the Natural Language Processing component of the AI Assistant and the Logical Parser will analyze the uploaded text to determine the structure and intent of the uploaded text. Uploaded text will be processed first by a number of analysis tools internally inside the AI Analysis Module, then it will be evaluated for meaning and structure via the AI Assistant's (with its Natural Language Processing capabilities) and Logical Parser.

V. RESULT AND DISCUSSION

The Intelligent Rubrics-Based Evaluator for Flowcharts, Algorithms, and Pseudocodes was tested using a set of different student submissions ranging from flowcharts to typed algorithms, to pseudocodes written in various styles. The evaluator was able to deal with various input formats. The OCR component functioned well to extract textual and structural details when dealing with image-based submissions. Though there was a minute loss in accuracy when it came to extractions from handwriting as opposed to typed submissions, pre-processing methods did contribute positively to improving readability and quality. The detection component for shapes inside a flowchart functioned well to determine standard symbols as well as control flow, allowing accurate determination of execution paths. Algorithm and pseudocode evaluation components had the ability to determine logical as well as syntactical sequences without depending on programming languages.

The rubric-assisted evaluation system enabled an unbiased grading process for all submissions made. Notably, manual assessment carried out by different assessors may yield different results. However, in this automated system, there was consistency in grading similar submissions made. Moreover, the feedback generated assisted students in learning from their mistakes in logic, steps, and clarity, as students received feedback on their submissions.

VI. CONCLUSION

The Intelligent Rubric-Based Evaluation System is a new type of assessment which automates the process of assessing student submissions by means of Artificial Intelligence and Predefined Evaluation Rubrics. This automated system is able to evaluate the submissions automatically for example flowcharts, algorithms, and written responses objectively, consistently, and in an organized manner.

The new Intelligent Rubric-Based Evaluation System integrates Optical Character Recognition (OCR), pre-trained transformer models associated with semantic reasoning guided by rubrics to create a much more efficient automated assessment process. This reduction of manual evaluation effort will help eliminate human bias in the method of evaluation.

The Intelligent Rubric-Based Evaluation System allows for the automatic assessment of both image-based and text-based submissions, giving the ability to identify the features of a submission to perform an assessment against an established set of criteria based on each submitted rubric. Because the established criteria within a rubric create a level of order, fairness will exist when grading. In addition, the Intelligent Rubric-Based Evaluation System will allow for faster, more precise assessments using AI technology.

Results from the current project demonstrate that the Intelligent Rubric-Based Evaluation System can provide consistent, automated assessments based on multiple

evaluation categories to meet the needs of educational institutions.

VII. FUTURE WORKS

While the current functionalities for the Proposed System are numerous, there remain many enhancements that could improve its utility. Some suggested enhancements included capabilities to Evaluate Program Code, Derive Mathematical Formulas, Visualise or draw Response(s) via Diagrams and the Implementation of Adaptive Rubrics. With regard to Adaptive Rubrics, these are likely to offer much better Evaluation(s) when creating an Evaluation of the answer to a specific question, than do the pre-defined criteria. An Adaptive Rubric would evaluate the Score given, based on how well the student answered the last several lists of questions and their response accuracy to each question within a specific List.

Some Additional enhancements for the Proposed System have been proposed, including Plagiarism Detection; Capability to support Multiple Languages through the use of Optical Character Recognition capability; and provides Real-Time Feedback to the Students who completed a Test. Through offering the opportunity for educators to be able to have a Solution that hosts a system such as this at Large Scale, within their Institutions using a Cloud-Based Solution, continuously refining the accuracy and reliability of how a Grading Model evaluates Student's Responses with expert input/review through the combination of Cloud-based Technology and Reinforcement Learning, will provide Educational Institutions with Efficient and Timely Feedback to Students.

REFERENCES

- [1] Ahmad Al-hafiz Sagala, & Yahfizham Yahfizham. (2024). Analisis Pengenalan Konsep Algoritma Pemrograman Matematika Pada Kehidupan Sehari Hari. *Morfologi : Jurnal Ilmu Pendidikan, Bahasa, Sastra Dan Budaya*, 2(1). <https://doi.org/10.61132/morfologi.v2i1.267>
- [2] Aiouni, R., Bey, A., & Bensebaa, T. (2018). EALGO: An automated assessment tool of flowchart programs for novices. *International Journal of Innovation and Learning*, 23(1). <https://doi.org/10.1504/IJIL.2018.088785>
- [3] Andrzejewska, M., & Stolińska, A. (2022). Do Structured Flowcharts Outperform Pseudocode? Evidence from Eye Movements. *IEEE Access*, 10. <https://doi.org/10.1109/ACCESS.2022.3230981>
- [4] Daviduck, B. (2000). Introduction to Programming in C ++ : Algorithms , Flowcharts and Pseudocode. *Deans Hill Systems*.
- [5] Ghosh, S., Pratihari, S., Chatterji, S., & Basu, A. (2023). Matching of hand-drawn flowchart, pseudocode, and english description using transfer learning. *Multimedia Tools and Applications*, 82(17). <https://doi.org/10.1007/s11042-023-14346-9>
- [6] Jogiyanto. (2018). Desain Algoritma Operasi Perkalian Matriks Menggunakan Metode Flowchart. *Jurnal Teknik Komputer Amik Bsi*, 1(1).
- [7] Nita, S., & Kartikawati, S. (2020). Analysis of the Impact Narrative Algorithm Method, Pseudocode and Flowchart Towards Students Understanding of the Programming Algorithm Courses. *IOP Conference Series: Materials Science and Engineering*, 835(1). <https://doi.org/10.1088/1757-899X/835/1/012044>
- [8] Pambuko, Z. B., Setiawan, A., Rahayu, H. S. E., Rusdijjati, R., Setiyo, M., Aman, M., & Praja, C. B. E. (2019). Determining an algorithm as the researcher's contribution: An alternative to performance-based research honorarium. *Journal of Physics: Conference Series*, 1402(2). <https://doi.org/10.1088/1742-6596/1402/2/022078>
- [9] Scanlan, D. A. (1988). Learner preference for using structured flowcharts vs. pseudocode when comprehending short, relatively complex algorithms: A summary analysis. *The Journal of Systems and Software*, 8(2). [https://doi.org/10.1016/0164-1212\(88\)90007-6](https://doi.org/10.1016/0164-1212(88)90007-6)
- [10] Tarsini, I., & Anggraeni, R. (2024). Explore flowchart and pseudocode concepts in algorithms and programming. *Indonesian Journal of Multidisciplinary Science*, 3(5). <https://doi.org/10.55324/ijoms.v3i5.807>